MSG No. 69.524 $Fm'm'm$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `m'm'm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `m'm'm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[1,7,11,13]`
2	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[2,8,12,14]`
3	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[3,5,9,15]`
4	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4,6,10,16]`
5	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[17,23,27,29]`
6	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[18,24,28,30]`
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[19,21,25,31]`
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[20,22,26,32]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[1,8,19,22]`
2	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[2,7,20,21]`
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[3,6,17,24]`
4	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[4,5,18,23]`
5	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,16,27,30]`
6	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[10,15,28,29]`
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[11,14,25,32]`
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[12,13,26,31]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,4,26,27]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,3,25,28]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[5,8,30,31]`
4	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[6,7,29,32]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,12,18,19]`
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,17,20]`
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[13,16,22,23]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[14,15,21,24]`

Table 6: Wyckoff site: `8f`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,13,22,26]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,14,21,25]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,15,24,28]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,16,23,27]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,9,18,30]`
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,10,17,29]`
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,11,20,32]`
8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[8,12,19,31]`

Table 7: Wyckoff site: `8g`, site symmetry: `2'm'm`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,5,8]`
2	$[-x, 0, 0]$	`[2,3,6,7]`
3	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,13,16]`
4	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[10,11,14,15]`
5	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,20,21,24]`
6	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	`[18,19,22,23]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,28,29,32]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	`[26,27,30,31]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'2'm`

No.	position	mapping
1	$[0, y, 0]$	`[1,4,6,7]`
2	$[0, -y, 0]$	`[2,3,5,8]`
3	$\left[0, y+\frac{1}{2}, \frac{1}{2}\right]$	`[9,12,14,15]`
4	$\left[0, \frac{1}{2}-y, \frac{1}{2}\right]$	`[10,11,13,16]`
5	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[17,20,22,23]`
6	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[18,19,21,24]`
7	$\left[\frac{1}{2}, y+\frac{1}{2}, 0\right]$	`[25,28,30,31]`

continued ...

Table 8

No.	position	mapping
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[26,27,29,32]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'm'2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,7,8]`
2	$[0,\ 0,\ -z]$	`[3,4,5,6]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,12,13,14]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[17,18,23,24]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[19,20,21,22]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[25,26,31,32]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[27,28,29,30]`

Table 10: Wyckoff site: `16j`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,26]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[2,25]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[3,28]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[4,27]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[5,30]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[6,29]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[7,32]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[8,31]`
9	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[9,18]`
10	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[10,17]`
11	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,20]`
12	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[12,19]`
13	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[13,22]`
14	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[14,21]`
15	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[15,24]`
16	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[16,23]`

Table 11: Wyckoff site: `16k`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,22]`
2	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	`[2,21]`
3	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[3,24]`
4	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	`[4,23]`
5	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[5,18]`
6	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	`[6,17]`

continued ...

Table 11

No.	position	mapping
7	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	[7,20]
8	$\left[\frac{1}{4}, -y, \frac{1}{4}\right]$	[8,19]
9	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[9,30]
10	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[10,29]
11	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[11,32]
12	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[12,31]
13	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[13,26]
14	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[14,25]
15	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[15,28]
16	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[16,27]

Table 12: Wyckoff site: 16l, site symmetry: 2'..

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,13]
2	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[2,14]
3	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[3,15]
4	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[4,16]
5	$\left[x, \frac{3}{4}, \frac{3}{4}\right]$	[5,9]
6	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[6,10]
7	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	[7,11]
8	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[8,12]
9	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[17,29]
10	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[18,30]
11	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[19,31]
12	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[20,32]
13	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[21,25]
14	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[22,26]
15	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[23,27]
16	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[24,28]

Table 13: Wyckoff site: 16m, site symmetry: m'..

No.	position	mapping
1	$[0, y, z]$	[1,7]
2	$[0, -y, z]$	[2,8]
3	$[0, -y, -z]$	[3,5]
4	$[0, y, -z]$	[4,6]
5	$\left[0, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9,15]
6	$\left[0, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[10,16]
7	$\left[0, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[11,13]
8	$\left[0, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[12,14]
9	$\left[\frac{1}{2}, y, z+\frac{1}{2}\right]$	[17,23]
10	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	[18,24]

continued ...

Table 13

No.	position	mapping
11	$\left[\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[19,21]
12	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	[20,22]
13	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[25,31]
14	$\left[\frac{1}{2}, \frac{1}{2}-y, z\right]$	[26,32]
15	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[27,29]
16	$\left[\frac{1}{2}, y+\frac{1}{2}, -z\right]$	[28,30]

Table 14: Wyckoff site: 16n, site symmetry: .m'.

No.	position	mapping
1	$[x, 0, z]$	[1,8]
2	$[-x, 0, z]$	[2,7]
3	$[-x, 0, -z]$	[3,6]
4	$[x, 0, -z]$	[4,5]
5	$\left[x, \frac{1}{2}, z+\frac{1}{2}\right]$	[9,16]
6	$\left[-x, \frac{1}{2}, z+\frac{1}{2}\right]$	[10,15]
7	$\left[-x, \frac{1}{2}, \frac{1}{2}-z\right]$	[11,14]
8	$\left[x, \frac{1}{2}, \frac{1}{2}-z\right]$	[12,13]
9	$\left[x+\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[17,24]
10	$\left[\frac{1}{2}-x, 0, z+\frac{1}{2}\right]$	[18,23]
11	$\left[\frac{1}{2}-x, 0, \frac{1}{2}-z\right]$	[19,22]
12	$\left[x+\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[20,21]
13	$\left[x+\frac{1}{2}, \frac{1}{2}, z\right]$	[25,32]
14	$\left[\frac{1}{2}-x, \frac{1}{2}, z\right]$	[26,31]
15	$\left[\frac{1}{2}-x, \frac{1}{2}, -z\right]$	[27,30]
16	$\left[x+\frac{1}{2}, \frac{1}{2}, -z\right]$	[28,29]

Table 15: Wyckoff site: 16o, site symmetry: ..m

No.	position	mapping
1	$[x, y, 0]$	[1,4]
2	$[-x, -y, 0]$	[2,3]
3	$[x, -y, 0]$	[5,8]
4	$[-x, y, 0]$	[6,7]
5	$\left[x, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,12]
6	$\left[-x, \frac{1}{2}-y, \frac{1}{2}\right]$	[10,11]
7	$\left[x, \frac{1}{2}-y, \frac{1}{2}\right]$	[13,16]
8	$\left[-x, y+\frac{1}{2}, \frac{1}{2}\right]$	[14,15]
9	$\left[x+\frac{1}{2}, y, \frac{1}{2}\right]$	[17,20]
10	$\left[\frac{1}{2}-x, -y, \frac{1}{2}\right]$	[18,19]
11	$\left[x+\frac{1}{2}, -y, \frac{1}{2}\right]$	[21,24]
12	$\left[\frac{1}{2}-x, y, \frac{1}{2}\right]$	[22,23]
13	$\left[x+\frac{1}{2}, y+\frac{1}{2}, 0\right]$	[25,28]
14	$\left[\frac{1}{2}-x, \frac{1}{2}-y, 0\right]$	[26,27]

continued ...

Table 15

No.	position	mapping
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[29,32]
16	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[30,31]

Table 16: Wyckoff site: 32p, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[-x,\ -y,\ -z]$	[3]
4	$[x,\ y,\ -z]$	[4]
5	$[x,\ -y,\ -z]$	[5]
6	$[-x,\ y,\ -z]$	[6]
7	$[-x,\ y,\ z]$	[7]
8	$[x,\ -y,\ z]$	[8]
9	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[10]
11	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12]
13	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[18]
19	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[19]
20	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[27]
28	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[28]
29	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[30]
31	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[31]
32	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[32]